DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Tutorial Sheet 11: **4th and 5th May 2023**

1. Let $f:\mathbb{R}^2\to\mathbb{R}$ be a continuous function and let D be a domain in $\mathbb{R}^2$ such that

$$\iint_D f(x,y)dxdy = \int_0^2\int_{y^3}^{y+6} f(x,y)dxdy\,. \qquad (*)$$

   (a) Sketch the region D.

   (b) Change the order of integration in $(*)$.

2. **(Change of variables in double integrals)**

$$\boxed{\iint_D f(x,y)dxdy = \iint_R f(x(s,t),y(s,t))\big|J_T(s,t)\big|dsdt} \qquad (**)$$

   where $T:\mathbb{R}^2\to\mathbb{R}^2$ is invertible with T, T^{-1} having continuous partial derivatives and Jacobian J_T different from 0 also $(x,y)=T(s,t)$.

   (a) Evaluate $\displaystyle\iint_D xydxdy$ where D is the parallelogram bounded by the lines $x-3y=1$, $x-3y=3$, $2x+y=-2$ and $2x+y=0$.

   (b) Let $D:=\{(x,y)\in\mathbb{R}^2\colon\ 0\le x\le 2,\quad 0\le x-y\le 2\}$.

   Evaluate the double integral $\displaystyle\iint_D \frac{(x-y)^2}{1+(x-y)^2}\,dxdy\,.$

3. **(Evaluation of double integral using polar coordinates)** We recall that if $T:\mathbb{R}^2\to\mathbb{R}^2$ is defined by $(x,y)=T(r,\theta)$ with $x=r\cos\theta$ and $y=r\sin\theta$ and $D=T(R)$ then

$$\boxed{\iint_D f(x,y)dxdy = \iint_R f(r\cos\theta,r\sin\theta)rdrd\theta}\,. \qquad (***)$$

   Evaluate each of the following integrals:

   (i) $\displaystyle\iint_D \frac{1}{1+x^2+y^2}dxdy$ where $D:=\{(x,y)\in\mathbb{R}^2\colon\ x^2+y^2\le 4,\ y\le 0\}$.

(ii) $\displaystyle\iint_D \frac{x}{x^2+y^2}dxdy$ where $D := \{(x,y) \in \mathbb{R}^2\colon\ 0 \le y \le x,\ x^2+y^2 \ge 1,\ x \le 1\}$.

4. **(After the lecture on Friday)** Evaluate the triple integral $\displaystyle\iiint_R 2y^2\sqrt{x}dxdydz$ where

$$R = \{(x,y,z)\colon\ -1 \le y \le 1,\, 0 \le x \le 1-y^2,\, -\sqrt{x} \le z \le \sqrt{x}\}$$

5. **(After the lecture on Friday)** Let $a > 0$, $b > 0$ and $c > 0$. If T is the tetrahedron bounded by the three coordinate planes and the plane $\dfrac{x}{a}+\dfrac{y}{b}+\dfrac{z}{c}=1$ then show that the volume of T is equal to $\dfrac{abc}{6}$.

6. **(After the lecture on Friday)** Convert the integral $\displaystyle\int_{-2}^{2}\int_{0}^{\sqrt{4-y^2}}\int_{0}^{x}(x^2+y^2)dzdxdy$ into an equivalent integral in cylindrical coordinates and evaluate the result.

7. **(After the lecture on Friday)** Let E be the region above the cone $z=\sqrt{x^2+y^2}$ and inside the sphere $x^2+y^2+z^2=4$. Evaluate the triple integral $\displaystyle\iint_E (x^2+y^2)zdxdydz$ through:

   (a) cylindrical coordinates;

   (b) spherical coordinates.